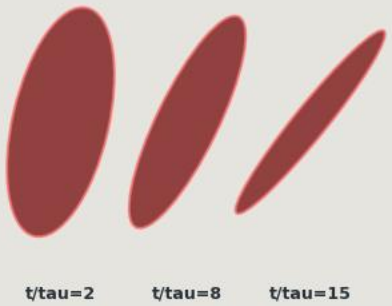
SHARED 3D VISUALIZATION ENGINE . TRACK 3 INTERFACIAL RHEOLOGY ADAPTATION

Interface-Resolved Emulsion Dynamics: From Constitutive Rheology to Multiphase Flow

Translate the wireframe-over-core motif to bitumen emulsion droplets under shear, temperature, and emulsifier kinetics.

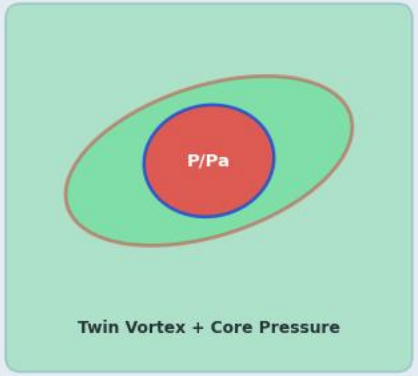
1 . DROPLET INTERFACE & DEFORMATION (t / tau_eta = 1..15)

* Bitumen droplet core + interface mesh
* Taylor deformation D = (L - B)/(L + B) under Ca = mu_c * gamma_dot * a0 / sigma
* Boussinesq-Scriven temperature-dependent interfacial viscosity
* Captures elongation & necking without biological release shell assumptions



2 . MULTIPHASE VORTEX FLOW & PRESSURE FIELD (P / Pa)

* Internal twin-vortex circulation and external shear streamlines (a, b, c)
* Hydrodynamic pressure P/Pa in [-0.30, 1.40] & emulsifier Gamma
* Validated against DSR & moisture-susceptibility measurements
* Track 3 rejects cytometry coupling



Architecture rule: reuse 3D wireframe/isosurface rendering and data infrastructure, not unvalidated physical parameters across tracks.

Track 3 Bitumen Emulsion & Interfacial Rheology: coupled constitutive rheology (Julia) + multiphase flow (OpenFOAM), separate from Track 1/Track 2.